

WHEN ARE TRIANGLES INVISIBLE? FUNDAMENTAL LIMITS OF HIGHER-ORDER STRUCTURE IDENTIFIABILITY FROM EDGE FLOWS

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ABSTRACT

Topological signal processing places signals on the triangles of a simplicial complex, but which triangles are “filled” is often unknown and must be inferred from edge flows. Existing work gives algorithms; we ask when the task is *possible at all*. The latent structure is observed only through the *curl* of the flows, which annihilates gradient and harmonic nuisances; classical detection theory then yields a curl-invisibility threshold $\rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$, a $\log \binom{p}{k}$ Fano converse for joint recovery, and exact per-triangle marginal laws with effective SNR $\sigma_c^2/(\sigma_n^2(\mathbf{G}^{-1})_{\tau\tau})$ after a GLS-type decorrelation. Our main result corrects the folklore “rank obstruction”: it is a *first- vs. second-order identifiability dichotomy*. Equal-curl-subspace supports are indistinguishable under deterministic signals — on K_n the curl subspace carries a $3/n$ degrees-of-freedom ratio — yet under random signals *every* support is identifiable at *any* rank deficiency, because the lifted signatures $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}$ are always linearly independent. Unfavorable geometry costs sample complexity, not possibility — and remarkably little: the worst-case equal-image confuser (adding a tetrahedron’s fourth face) has exponent $\frac{9}{16}\rho_2^2(1+o(1))$, $\rho_2 = \sigma_c^2/\sigma_n^2$, *exactly* the isolated-triangle detection exponent — rank deficiency is free at the exponent level. All constants are exhaustively and Monte-Carlo validated, and the framework recovers genuine, unplanted structure from real data: tropical-cyclone circulation from ERA5 edge flows, validated against source-grid vorticity and the independent IBTrACS archive. Code and data are released.

Index Terms— Topological signal processing, simplicial complexes, Hodge decomposition, identifiability, detection theory, edge flows.

1. INTRODUCTION

Graph signal processing captures pairwise structure, but many networked measurements—currency exchange, traffic circulation, neural flow, statistical ranking—carry *higher-order* structure that lives on the triangles of a simplicial complex [1, 2, 3]. Topological signal processing (TSP) has rapidly built a toolbox on the Hodge Laplacian: convolutional filters [4], maximally localized dictionaries [5], Dirac operators [6], adaptive estimation [7], and unrolled reconstruction [8]. A recurring primitive is that the set of filled triangles is unknown and must be learned from edge-flow signals.

Prior work. Learning a graph topology from node signals is well charted [9, 10]. Its higher-order analogue — inferring which triangles are filled — has been treated *algorithmically*: MAP estimation [11], greedy topology learning [12], sparse cell-complex augmentation [13]; adaptive schemes [7] list structure discovery

as open. HodgeRank [14] established the curl as the signature of higher-order inconsistency. Missing is the *converse side*: when is the task information-theoretically solvable, at what snapshot budget, and how does the answer depend on the geometry? (Simplicial/hypergraph SBM detectability observes the structure *directly*; here it is latent, seen only through the flows it shapes.) We supply that converse side. Our tools are deliberately classical — detection theory and Fano arguments [15, 16], in the lineage of support-recovery and sparse-covariance detection limits [17, 18, 19, 20] — our contribution is the reduction that makes them bite, plus one dichotomy theorem (Thm. 3) we believe is genuinely new:

1. A *curl-annihilation* reduction (Lemma 1; the detection-theoretic use of HodgeRank’s observation): the latent structure is observable only through the curl, giving a standard two-variance test (Prop. 1) and an *exact* finite-sample product law over edge-disjoint candidates (Prop. 2).
2. Converse thresholds through the reduction (Thm. 1): the sample-complexity scaling $\rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ at the exponent level, and a Fano $\log \binom{p}{k}$ factor for joint recovery.
3. *Geometry-aware decorrelation* for edge-sharing candidates — the GLS/BUE inversion $\hat{\mathbf{y}} = \mathbf{G}^{-1}\mathbf{c}$ [21] — with exact marginal laws whose effective SNR is the variance-inflation factor’s (Thm. 2), plus a union-bound guarantee (Cor. 1).
4. Main theorem: a *first- vs. second-order identifiability dichotomy* (Thm. 3) correcting the folklore rank obstruction: equal-image supports are indistinguishable under deterministic signals (on K_n : a $3/n$ DoF ratio), yet under random signals *every* support is identifiable at *any* rank deficiency (lifted-spark Lemma 2) — and the worst-case price *equals the isolated-triangle exponent* $\frac{9}{16}\rho_2^2$: rank deficiency is free at the exponent level.

Every result is Monte-Carlo validated. On real data we recover *genuine, unplanted* structure — ERA5 cyclone circulation, validated against source-grid vorticity and the independent IBTrACS archive — with FX as a consistency check and road networks grounding the recovery laws.

2. BACKGROUND AND SIGNAL MODEL

Fix a graph $(\mathcal{V}, \mathcal{E})$ with node–edge incidence $\mathbf{B}_1 \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$ and, for a candidate triangle set $\mathcal{T}$ (e.g. all 3-cliques), edge–triangle incidence $\mathbf{B}_2 \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{T}|}$; N is the snapshot count, $p = |\mathcal{T}|$. The identity $\mathbf{B}_1 \mathbf{B}_2 = \mathbf{0}$ holds, and the Hodge 1-Laplacian $\mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top$ induces the orthogonal split

$$\mathbb{R}^{|\mathcal{E}|} = \underbrace{\text{im } \mathbf{B}_1^\top}_{\text{gradient}} \oplus \underbrace{\text{im } \mathbf{B}_2}_{\text{curl}} \oplus \underbrace{\ker \mathbf{L}_1}_{\text{harmonic}}. \quad (1)$$

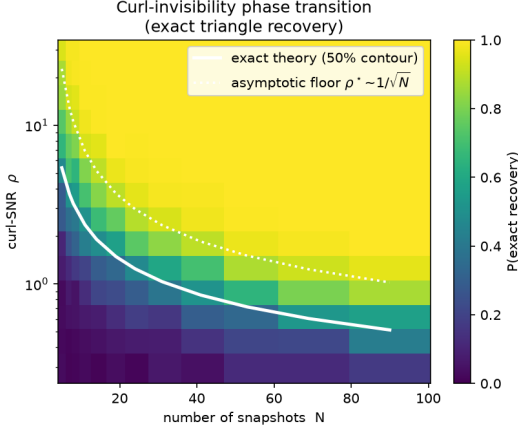


Fig. 1. Curl-invisibility recovery threshold. Colour is empirical $P(\text{exact recovery})$; the solid line is the exact finite-sample law (Prop. 2), the dotted line the Bonferroni-corrected Chernoff floor of Thm. 1 ($\delta \rightarrow \delta/p$).

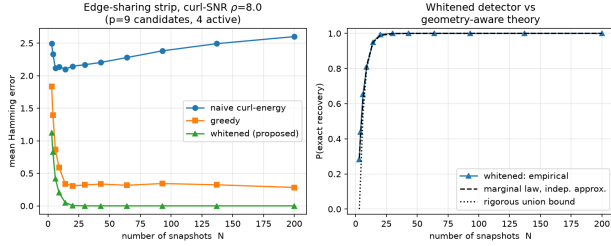


Fig. 2. Edge-sharing strip. Left: mean Hamming error vs. N ; the naive detector never recovers. Right: decorrelated detector vs. the exact marginal theory — independence approximation (dashed) and rigorous union bound (dotted).

The *curl* of a flow is $\text{curl } \mathbf{f} = \mathbf{B}_2^\top \mathbf{f}$. Each column of $\mathbf{B}_2$ has three ± 1 entries, so $\mathbf{G} = \mathbf{B}_2^\top \mathbf{B}_2$ has $G_{\tau\tau} = 3$, $G_{\sigma\tau} \in \{0, \pm 1\}$, nonzero iff σ, τ share an edge. The 1-skeleton is known; the latent object is the filled set $\mathcal{S} \subseteq \mathcal{T}$. We observe N i.i.d. snapshots

$$\mathbf{f}_t = \mathbf{B}_1^\top \mathbf{a}_t + \mathbf{B}_{2,\mathcal{S}} \mathbf{y}_t + \mathbf{h}_t + \mathbf{n}_t, \quad t = 1, \dots, N, \quad (2)$$

with $\mathbf{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I})$ (gradient nuisance), $\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, \sigma_c^2 \mathbf{I})$ on $\mathcal{S}$, harmonic $\mathbf{h}_t \in \ker \mathbf{L}_1$, noise $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$; the goal is to recover $\mathcal{S}$. This is the generative family of the algorithmic literature (cf. [11]); we characterize when its inverse problem is solvable.

3. FUNDAMENTAL LIMITS

3.1. Curl annihilation and the single-triangle test

Lemma 1 (Curl annihilation). *For any $\mathbf{f}$, the statistic $\mathbf{B}_2^\top \mathbf{f}$ depends only on the curl component: $\mathbf{B}_2^\top (\mathbf{B}_1^\top \mathbf{a}) = \mathbf{0}$ for all $\mathbf{a}$ and $\mathbf{B}_2^\top \mathbf{h} = \mathbf{0}$ for all $\mathbf{h} \in \ker \mathbf{L}_1$. Under (2),*

$$\mathbf{c}_t = \mathbf{B}_2^\top \mathbf{f}_t = (\mathbf{B}_2^\top \mathbf{B}_{2,\mathcal{S}}) \mathbf{y}_t + \mathbf{B}_2^\top \mathbf{n}_t. \quad (3)$$

Proof. $\mathbf{B}_1 \mathbf{B}_2 = \mathbf{0}$ gives $\mathbf{B}_2^\top \mathbf{B}_1^\top = \mathbf{0}$. If $\mathbf{L}_1 \mathbf{h} = \mathbf{0}$ then $0 = \mathbf{h}^\top \mathbf{L}_1 \mathbf{h} \geq \mathbf{h}^\top \mathbf{B}_2 \mathbf{B}_2^\top \mathbf{h} = \|\mathbf{B}_2^\top \mathbf{h}\|^2$, so $\mathbf{B}_2^\top \mathbf{h} = \mathbf{0}$. $\square$

Thus the nuisances—however large—are removed exactly; all information about $\mathcal{S}$ flows through (3). For an *isolated* candidate τ

(no shared edges) the scalar $c_{\tau,t}$ is zero-mean Gaussian with variance $v_0 = 3\sigma_n^2$ if $\tau \notin \mathcal{S}$ and $v_1 = 9\sigma_c^2 + 3\sigma_n^2 = v_0(1 + \rho)$ if $\tau \in \mathcal{S}$, where

$$\rho \triangleq 3\sigma_c^2/\sigma_n^2 \quad (4)$$

is the *curl signal-to-noise ratio*.

Proposition 1 (Two-variance detection). *Detecting an isolated τ from N snapshots is a standard two-variance Gaussian test [15, 16]: the energy $E = \sum_{t=1}^N c_{\tau,t}^2$ is sufficient, $E/v_i \sim \chi_N^2$ under H_i , the minimum Bayes error is attained by thresholding E , admits a closed form via χ_N^2 tails, and decays as $\exp(-NC(v_0, v_1))$ with C the Gaussian Chernoff information $\max_s \frac{1}{2} [\log(s/v_0 + (1-s)/v_1) + s \log v_0 + (1-s) \log v_1]$.*

Theorem 1 (Curl-invisibility threshold; joint-recovery converse). (a) With ρ as in (4), $C(\rho) = \rho^2/16 + o(\rho^2)$; hence attaining a target error exponent δ at budget N requires $\rho \geq \rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ for any estimator. (b) Recovering a uniformly random size- k support among p edge-disjoint candidates with error $\leq \epsilon$ requires $N \geq ((1-\epsilon) \log \binom{p}{k} - \log 2)/(k \text{KL}_1)$, $\text{KL}_1 = (\rho - \log(1+\rho))/2$; joint recovery pays a $\log \binom{p}{k}$ factor.

Proof sketch. (a) Chernoff information is the optimal Bayes error exponent for i.i.d. observations [16]; Taylor expansion gives $\rho^2/16$, and $e^{-NC(\rho)} \leq \delta$ forces the scaling. (b) Independent curl coordinates give $I(\mathcal{S}; \mathbf{c}^N) \leq Nk \text{KL}(\mathcal{N}(0, v_1) \parallel \mathcal{N}(0, v_0))$; Fano finishes (supplement §S1, incl. a signal-agnostic variant). Honest scope: (a) is exponent-level — the finite- N 50% boundary sits an order-one factor below the floor (same $1/\sqrt{N}$ law, Fig. 1); the scaling is the classical variance-detection rate [15, 17]; specific to our problem are the reduction producing ρ and the constants (3, 9). $\square$

When the candidates are pairwise edge-disjoint, the coordinates of $\mathbf{c}_t$ are *independent* across triangles, so support recovery decouples exactly:

Proposition 2 (Exact finite-sample recovery law). *For edge-disjoint candidates and the per-triangle Bayes-optimal energy threshold, $P(\hat{\mathcal{S}} = \mathcal{S}) = \prod_{\tau \in \mathcal{S}} (1 - P_{\text{miss}}) \prod_{\tau \notin \mathcal{S}} (1 - P_{\text{fa}})$, with $P_{\text{miss}}, P_{\text{fa}}$ given by χ_N^2 tails at the common threshold. Its 50% contour in the (N, ρ) plane is the empirical recovery boundary (Fig. 1).*

3.2. Confusable triangles: geometry-aware decorrelation

Edge-sharing candidates are *confusable*: an active triangle leaks curl energy onto neighbours through the off-diagonal entries of $\mathbf{G}$, so naive energy thresholding fails (Fig. 2, left) and *worsens* as ρ grows. The repair is the textbook GLS/BLUE inversion of the curl map [21] (for invertible $\mathbf{G}$, OLS/GLS/BLUE/ML all coincide at $\mathbf{G}^{-1}\mathbf{c}$); what matters is the *exact marginal detection law* it induces, whose geometry factor is the classical variance-inflation factor.

Theorem 2 (Decorrelated marginal laws). *Assume $\mathbf{B}_2$ has full column rank, so $\mathbf{G}$ is invertible. Let $\hat{\mathbf{y}}_t = \mathbf{G}^{-1}\mathbf{c}_t$ and let $\tilde{\mathbf{y}}_t$ denote $\mathbf{y}_t$ zero-padded to all of $\mathcal{T}$. Then, exactly,*

$$\hat{\mathbf{y}}_t = \tilde{\mathbf{y}}_t + \mathbf{e}_t, \quad \mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{G}^{-1}). \quad (5)$$

Hence the marginal law of each coordinate $\hat{y}_{\tau,t}$ is a two-variance test with $v_{0,\tau} = \sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau}$, $v_{1,\tau} = \sigma_c^2 + v_{0,\tau}$, and per-triangle effective SNR

$$\rho_\tau^{\text{eff}} = \frac{\sigma_c^2}{\sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau}}. \quad (6)$$

Proof. By Lemma 1, $\mathbf{c}_t = \mathbf{G}\tilde{\mathbf{y}}_t + \mathbf{B}_2^\top \mathbf{n}_t$ (the columns of $\mathbf{G}$ indexed by $\mathcal{S}$ act on $\mathbf{y}_t$; zero-padding absorbs the indexing). Applying $\mathbf{G}^{-1}$ gives the mean exactly; the noise covariance is $\sigma_n^2 \mathbf{G}^{-1} \mathbf{G} \mathbf{G}^{-1} = \sigma_n^2 \mathbf{G}^{-1}$. Gaussian marginals of (5) give the per-coordinate two-variance laws. $\square$

$\mathbf{G}^{-1}$ decorrelates the *mean* (removing leakage exactly); the noise stays correlated — $\sigma_n^2 \mathbf{G}^{-1}$ is not diagonal — hence no “whitening”, and the joint law does *not* factorize. $(\mathbf{G}^{-1})_{\tau\tau}$ is the price of geometry: $1/3$ when isolated (recovering $\rho_\tau^{\text{eff}} = \rho$), growing as signatures overlap. What survives arbitrary correlations is a union bound over exact marginals:

Corollary 1 (Rigorous recovery guarantee). *Threshold each $\sum_t \hat{\mathbf{y}}_{\tau,t}^2$ at its marginal-optimal (or Bonferroni) level; with $P_{\text{err},\tau}$ the marginal errors of Theorem 2, $\Pr[\hat{\mathcal{S}} = \mathcal{S}] \geq 1 - \sum_\tau P_{\text{err},\tau}$, regardless of the correlation of $\mathbf{e}_t$.*

Remark 1 (Independence approximation). *The product $\prod_\tau (1 - P_{\text{err},\tau})$ is exact for diagonal $\mathbf{G}$ (Prop. 2) and an approximation otherwise; on our diagonally-dominant benchmarks it tracks Monte-Carlo within 0.03, the union bound remaining a valid, more conservative floor (Fig. 2, right).*

3.3. The rank obstruction is a dichotomy, not impossibility

When curl signatures are linearly dependent ($\mathbf{G}$ singular — e.g. all $\binom{n}{3}$ triangles of K_n), the folklore answer — “identifiable only modulo $\ker \mathbf{B}_2$ ” — is correct for *deterministic* signals but *false* for the random model (2): the covariance carries $\sigma_c^2 \sum_{\tau \in \mathcal{S}} \mathbf{b}_\tau \mathbf{b}_\tau^\top$, strictly finer than the column space. The correct statement is a dichotomy, resting on a spark property of the *lifted* signatures:

Lemma 2 (Lifted spark). *For any candidate set of distinct 3-cliques of a simple graph, the matrices $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. A pair of distinct edges lies in at most one common triangle, so for $e \neq e'$ both in τ , the (e, e') entry of $\sum_\tau w_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top$ equals $\pm w_\tau$: every coefficient is read off a distinct matrix entry. $\square$

Theorem 3 (First- vs. second-order identifiability). *Let $r = \text{rank } \mathbf{B}_2$ and write $\rho_2 = \sigma_c^2 / \sigma_n^2$. (i) First order. If $\text{im } \mathbf{B}_{2,\mathcal{S}} = \text{im } \mathbf{B}_{2,\mathcal{S}'}$, then under unknown deterministic triangle signals the two supports induce identical families of flow distributions — indistinguishable at any SNR and any N . Such confusers exist whenever the candidate set contains all four faces of a tetrahedron ($\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$); on K_n , $\dim \text{im } \mathbf{B}_2 = \binom{n-1}{2}$, a $3/n$ DoF ratio. (ii) Second order. Under the random model (2), the map $\mathcal{S} \mapsto \text{Cov}(\mathbf{f})$ is injective (Lemma 2), so every support is identifiable from i.i.d. snapshots — at any rank deficiency, including all of K_n . (iii) The price. The equal-image confusers of a hosted tetrahedron come in two kinds, with exact curl-domain separations: the face swap $\mathcal{S}' = \mathcal{S} \setminus \{\tau_a\} \cup \{\tau_b\}$ has $\|\mathbf{u}_a \mathbf{u}_a^\top - \mathbf{u}_b \mathbf{u}_b^\top\|_F^2 = 9 + 9 - 2 = 16$ (minimal among supports of equal cardinality; exhaustive on K_5), while the subset confuser $\mathcal{S}' = \mathcal{S} \cup \{\tau_4\}$ (add the fourth face) has $\|\mathbf{u}_4 \mathbf{u}_4^\top\|_F^2 = 9$ — the unrestricted worst case (exhaustive on K_5). The Bhattacharyya expansion $C_G = \frac{1}{16} \|\Delta M\|_F^2 \rho_2^2 (1 + o(1))$ gives exponents ρ_2^2 and $\frac{9}{16} \rho_2^2$ respectively — and $\frac{9}{16} \rho_2^2$ equals the isolated-triangle exponent $C(\rho) = \rho^2/16$ at $\rho = 3\rho_2$: at the exponent level, the hardest equal-image decision costs no more than detecting one isolated triangle, so rank deficiency is free and only the Fano log-multiplicity over the $4\binom{n}{4}$ leave-one-face-out hypotheses ($\asymp \log n$ on K_n) reflects the geometry.*

Proof sketch. (i) Equal images make the deterministic mean families coincide. (ii) In curl coordinates $\mathbf{z}_t = \mathbf{Q}^\top \mathbf{f}_t$ ($\mathbf{Q}$ orthonormal basis of $\text{im } \mathbf{B}_2$) the law is $\mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I}_r + \sigma_c^2 \sum_{\tau \in \mathcal{S}} \mathbf{u}_\tau \mathbf{u}_\tau^\top)$, $\mathbf{u}_\tau = \mathbf{Q}^\top \mathbf{b}_\tau$; injectivity transfers from Lemma 2 since $\mathbf{b}_\tau \in \text{im } \mathbf{B}_2$. (iii) Direct computation plus the second-order (Bhattacharyya) expansion of the Chernoff information; every constant is verified in the released tests: swap $C_G/\rho_2^2 \rightarrow 1$ (to 0.998), subset $C_G \rightarrow \frac{9}{16} \rho_2^2$ matching the single-triangle exponent to 10^{-4} relative, and the separations 16 (equal-cardinality minimum) and 9 (unrestricted minimum) are exhaustive over all 2^{10} supports of K_5 (test_exhaustive_k5_confuser_separations). Full proofs: supplement §S4. $\square$

Theorem 3 reconciles theory with practice: second-order estimators — fitting the empirical curl covariance with the atoms $\{\mathbf{g}_\tau \mathbf{g}_\tau^\top\}$ — achieve regime (ii), verified on rank-deficient K_5 ; the $3/n$ ratio survives as *first-order* observability.

Remark 2 (Open). *Sharp first-order sparse-support thresholds over $\ker \mathbf{B}_2$ (spark/coherence conditions) and the exact constant in (iii)’s log n factor remain open.*

4. EXPERIMENTS

All experiments run on CPU from fixed seeds; every theoretical curve is overlaid on independent Monte-Carlo simulation, and a released test-suite checks each closed form against simulation. Code and data reproduce all figures via a single script.

4.1. Recovery threshold (validation of Thm. 1, Prop. 2)

Fig. 1 sweeps (N, ρ) on an edge-disjoint planted complex ($p = 8$, 4 active, gradient/harmonic nuisance present). Recovery transitions rapidly (the steepness is the finite- p product of sigmoids — no critical-point claim); the exact product law of Prop. 2 tracks the empirical 50% contour to a median ratio 1.01; the dotted Bonferroni-corrected Chernoff floor ($\delta \rightarrow \delta/p$) shows the $1/\sqrt{N}$ law.

4.2. Confusability (validation of Thm. 2, Cor. 1)

On an edge-sharing strip ($p = 9$, alternating 4 active, $\rho = 8$; full column rank), the naive energy detector plateaus at Hamming error ≈ 2.6 regardless of N , while the decorrelated detector of Theorem 2 recovers exactly (Fig. 2, left); its recovery probability tracks the independence approximation within Monte-Carlo error and respects the union bound, conservative in the transition as a bound should be (right). A greedy residual-fitting baseline (standing in for [12]) stalls at ≈ 0.3 .

4.3. Second-order identifiability where subspaces fail (Thm. 3)

On K_5 ($p = 10$, $\text{rank } \mathbf{B}_2 = 6$: where a subspace argument forbids recovery), the greedy covariance-atom estimator recovers a planted 4-triangle support exactly ($N = 1200$, gradient nuisance present). The Monte-Carlo error of the optimal confuser-pair test respects the e^{-NC_G} envelope with empirical exponent converging onto C_G ; all separations and both exponent constants are verified as in Thm. 3(iii).

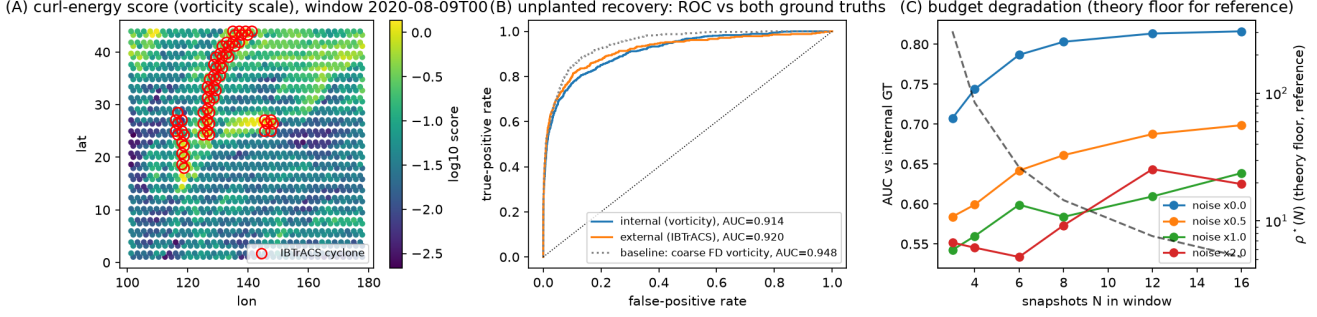


Fig. 3. Unplanted recovery of real higher-order structure. (A) Temporally centered curl-energy scores (vorticity scale) on the mesh for one 4-day window; red circles: independent IBTrACS cyclone fixes. (B) ROC across all windows against the same-field vorticity reference and the independent IBTrACS labels, with the classical coarse-vorticity baseline. (C) Detection quality vs. snapshot budget N (theory floor $\rho^*(N)$ for reference).

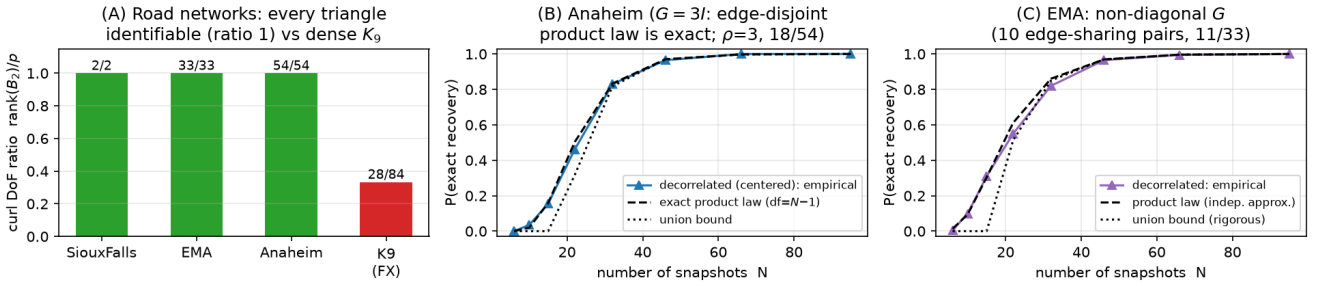


Fig. 4. Real road-network geometry. (A) Full-column-rank B_2 : DoF ratio 1 vs. K_9 's $3/n$. (B) Anaheim ($G = 3I$): planted recovery follows the exact edge-disjoint product law (df $N-1$). (C) EMA (non-diagonal G): heterogeneous marginal laws on real geometry — empirical vs. independence approximation and union bound.

4.4. Real, unplanted recovery: cyclones from wind-field edge flows

The achievability side is exercised on *genuine* latent structure (Fig. 3): tropical-cyclone circulation in ERA5 winds [22], Western North Pacific, Aug–Sep 2020. Ten-meter winds (0.7° grid) become edge flows on a triangulated mesh; by Stokes' theorem a triangle's curl is its circulation, so high-curl triangles are real atmospheric vortices. The mesh is simply connected, so B_2 has full column rank by Euler's formula: no confusers, the favorable regime of Theorem 3. Per 4-day window (16 six-hourly snapshots) the temporally centered curl-energy statistic of Prop. 1, area-normalized to the vorticity scale, scores each triangle; *nothing is planted, no oracle parameters* (the decorrelated variant, reported in the repository, ranks worse here: G^{-1} amplifies noise). Storms translate several triangle-widths per window: centering removes the window-mean flow, so the statistic detects the *time-varying* circulation a moving vortex deposits. Validation (panel B): a *same-field consistency reference* (finite-difference vorticity on the $3\times$ -finer source grid — a different functional of the same winds) and the genuinely independent IBTrACS archive [23]. Against a pointwise-vorticity baseline at the same information budget, the edge-flow statistic is better on the independent external reference (0.920 vs. 0.898), slightly worse on the internal one (0.914 vs. 0.948), which shares the baseline's functional — the contribution is the limits framework, not a new cyclone detector. Detection degrades as the budget shrinks (panel C; theory floor for reference).

4.5. Geometry on real infrastructure; FX as a consistency check

Fig. 4 grounds the recovery laws on real road geometry (TNTF [24]). (A) All three networks have full-column-rank B_2 (DoF ratio 1 vs. $3/n$ on K_9); the real equilibrium flows carry genuine rotational energy (2.4%/4.6% curl). (B) Anaheim ($G = 3I$), so planted recovery *collapses to the product law* (Prop. 2): the constant real background is centered out exactly (df $N-1$), the nuisances annihilated — stated plainly, the surviving statistical problem is synthetic; the honest real recovery evidence is Fig. 3. (C) EMA exercises Theorem 2 on real geometry (10 edge-sharing pairs, non-diagonal G): empirical recovery tracks the independence approximation and respects the union bound. *FX consistency check* (no figure): ECB-rate edge flows on K_9 are by construction a single-potential gradient $\mathbf{f} = B_1^\top \mathbf{p}$, so Lemma 1 predicts identically zero curl; measured curl/gradient = 1.8×10^{-31} — floating-point residue, a consistency check (HodgeRank's arbitrage-free reading [14]), not a market discovery. K_9 's 28/84 rank is panel (A)'s red bar.

5. CONCLUSION

We characterized when higher-order structure is identifiable from edge flows: a curl-invisibility threshold with a $\log \binom{p}{k}$ joint converse; exact marginal laws after GLS-type decorrelation; and a first- vs. second-order dichotomy whose worst-case price equals one isolated-triangle detection — delimiting when algorithmic TSP inference [11, 12, 13] succeeds, with achievability shown on unplanted ERA5 cyclone circulation. The supplement adds edge-sampling laws [25].

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